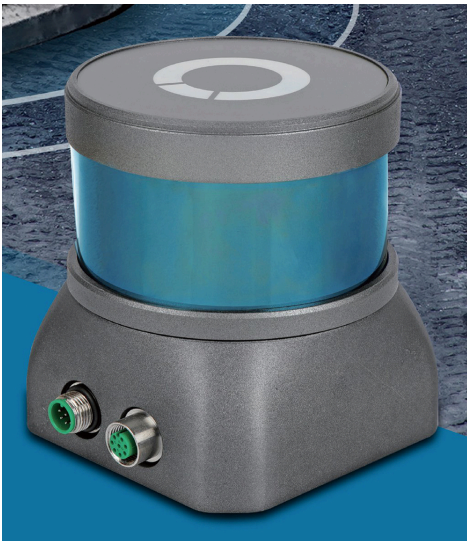


INTRINSICALLY SAFE 3D LIDAR · LR-16F-IS

INTRINSICALLY OLEI LR-16F-IS SAFE 3D LIDAR

The LR-16F optical core in an intrinsically safe, mine-rated enclosure: Ex ib I Mb, 100 metres at 80% reflectivity, 16 lines, three isolated power rails and twin 8-pin aviation plugs, for roadway modelling and mobile-equipment perception underground.



■ WHY THIS SENSOR

- **Ex ib I Mb intrinsically safe construction** — with a mining product safety sign and explosion-proof conformity certificate — see the approval note overleaf
- **100 m working distance at 80% reflectivity** — 16 lines across $\pm 15^\circ$ vertical, full 360° horizontal
- **310 000 points per second** — 0.18° horizontal at 10 Hz, 2° vertical spacing
- **$\leq \pm 5$ cm absolute, $\leq \pm 3$ cm repeat accuracy** — range resolution 2 mm, reflectivity data across 5–100%
- **High reflector recognition** — reflective survey targets picked out of the cloud automatically
- **Three isolated power rails** — 8–14 V, 14–17 V and 14–17 V, ≤ 11 W total, on twin 8-pin aviation plugs
- **IP66, -10 to $+50^\circ\text{C}$, < 1500 g** — 96 mm diameter $\times$ 110 mm, aluminium and PC construction
- **Built for roadways and working faces** — 3D modelling, mobile-equipment obstacle avoidance, roadway deformation and slope displacement monitoring, coal measurement

Ex ib I Mb

MINE-USED IS RATING

100 m

AT 80% REFLECTIVITY

16

LASER CHANNELS

310K

POINTS / SECOND

■ APPLICATION SCENARIOS



MINING & ROADWAY

Roadway modelling, slope and deformation monitoring



TUNNEL & UNDERGROUND

Profile capture in narrow, GPS-denied workings



INDUSTRIAL AUTOMATION

Guarded cells, robot envelopes and access ways



AMR & AGV

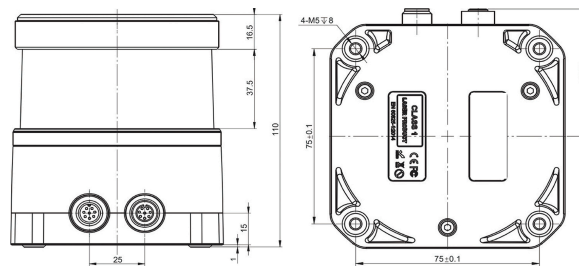
Full-circle obstacle avoidance from one chassis mount

LR-16F-IS — SPECIFICATIONS

PARAMETER	TECHNICAL DATA
Explosion-protection marking	Ex ib I Mb
Safety sign no.	MFB210030
Laser wavelength	905 nm
Laser safety level	Class 1 (IEC 60825-1:2014)
Laser channels	16
Returns	Single (strongest, first)
Beam divergence	3 mrad
Field of view (horizontal)	360°
Field of view (vertical)	-15° to +15°
Scanning frequency	5 or 10 Hz
Angle resolution (vertical)	2° (no 0° line) ±0.1°
Angle resolution (horizontal)	0.18° ±10% @ 10 Hz
Working distance	100 m @ 80% white board
Range resolution ratio	2 mm
Accuracy	Absolute ≤ ±5 cm Repeat ≤ ±3 cm
Point cloud density	About 310 000 pts
Reflectivity data	5-100%
High reflector recognition	Yes

PARAMETER	TECHNICAL DATA
Ethernet connection	100M Ethernet
UDP packets contain	Distances · calibrated reflectivities Rotation angles · synchronised time stamps
Connection	2 × 8-pin aviation plugs
Mechanical dimensions	96 (D) × 110 (H) mm
Operating voltage	Power 1: 8-14 V Power 2: 14-17 V · Power 3: 14-17 V
Power consumption	≤ 11 W
Weight	< 1500 g
Degree of protection	IP66
Operating temperature	-10 to +50 °C
Storage temperature	-20 to +60 °C
Ambient light limit	< 80 000 lux
Air humidity	< 95% (non-condensing)
Mounting	Bottom

■ OUTLINE AND MOUNTING



UNIT: MM · DRAWINGS NOT TO SCALE

Certification. Explosion protection Ex ib I Mb; mining product safety sign MFB210030; explosion-proof conformity certificate issued by the China Mining Product Safety Approval and Certification Center Co., Ltd. Laser safety Class 1 (IEC 60825-1:2014). EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Hazardous-location approval — read before quoting. This product's explosion-protection marking is **Ex ib I Mb** with a mining product safety sign (MFB210030) and an explosion-proof conformity certificate issued by the China Mining Product Safety Approval and Certification Center. That is a **Chinese coal-mine approval**. It is **not** ATEX, not IECEx, not a North American Class I / II Division 1 or 2 listing, and not an MSHA approval. Do not represent it as approved for hazardous locations in the United States or Canada. US underground coal applications require MSHA approval; other North American classified areas require a listing from an OSHA-recognised NRTL. Confirm the intended jurisdiction and approval route with MorpheusTEK before quoting.

Notes. The 100 m figure is stated against an 80% reflectivity white board. Vertical spacing is 2° with no 0° line. Horizontal resolution is stated at 10 Hz. The three power rails are separate intrinsically safe circuits with their own voltage, current, capacitance and inductance limits — the barrier or IS power supply must be selected against those parameters, which are printed on the device label and listed in the manufacturer's certificate. Note also that OLEI's own page-2 header for this datasheet reads "Radar Level Sensor"; the product is a LiDAR.

Installation. Bottom mounting. Install, earth and power the unit strictly in accordance with the intrinsic-safety parameters on the device label and the governing installation standard for the jurisdiction — IS wiring, barriers and segregation are part of the approval, not accessories to it. A 360° sensor needs its full circle clear of structure. Clean the optical window on a schedule; coal dust accumulation shortens effective range before it shows as an error.

OLEI and the OLEI logo are trademarks of Hangzhou OLE-Systems Co., Ltd. MorpheusTEK is the exclusive North American distributor for OLEI LiDAR. Specifications are subject to change without notice; confirm current values with MorpheusTEK before design-in. Doc. MTK-OLEI-LR16FIS-2026.09 · Source: OLEI Multi-Line Mine-used Intrinsic Safety LiDAR datasheet, DOC.# DSEN-16FIS-202304.